Assignment No 13:

Write a simple program in SCALA using Apache Spark framework

Objective:

To understand the basics of Apache Spark framework.

To write and execute a simple Scala program using Spark.

To perform basic data processing using Spark (Word Count).

Prerequisites:

Basic knowledge of programming concepts

Familiarity with Scala language

Understanding of big data concepts

Basic command-line usage

Software Requirements:

Apache Spark

Scala

Java JDK (8 or above)

IDE (IntelliJ IDEA / Eclipse)

Theory:

Apache Spark is an open-source distributed computing framework for big data processing.

It uses in-memory computation for faster processing.

Word Count is a basic example demonstrating transformations and actions in Spark.

Sample Program:

```
import org.apache.spark.sql.SparkSession

object WordCountExample {
  def main(args: Array[String]): Unit = {
```

```
    val spark = SparkSession.builder()
      .appName("Word Count Example")
      .master("local[*]")
      .getOrCreate()

    val input = spark.read.textFile("input.txt")

    val wordCount = input
      .flatMap(line => line.split(" "))
      .groupByKey(word => word)
      .count()

    wordCount.show()
    spark.stop()
  }
}
```

Sample Input

Hello Spark
Hello Scala
Spark is fast

Sample Output

```
+-----+-----+
|value|count|
+-----+-----+
|Hello|    2|
|Spark|    2|
|Scala|    1|
|is   |    1|
|fast |    1|
+-----+-----+
```

Installation Steps

Step 1: Install Java

- Download and install JDK (8 or above)
- Set JAVA_HOME environment variable

Step 2: Install Scala

- Download Scala from official website
- Set SCALA_HOME and update PATH

Step 3: Install Apache Spark

- Download Spark pre-built package
- Extract the folder
- Set SPARK_HOME and update PATH

Step 4: Verify Installation
Open terminal and run:

```
spark-shell
```
If Spark shell starts successfully, installation is complete.

Execution Steps

1. Create a text file named input.txt
2. Write sample content in it
3. Compile Scala program
4. Run using:

```
spark-submit --class WordCountExample your-jar-file.jar
```

Conclusion

- Successfully implemented a simple Spark application using Scala.
- Learned how to process data using transformations and actions.
- Understood basic working of distributed data processing using Spark.

Viva Questions
1. What is Apache Spark?

2. What are transformations and actions?

3. Why is Scala used in Spark?